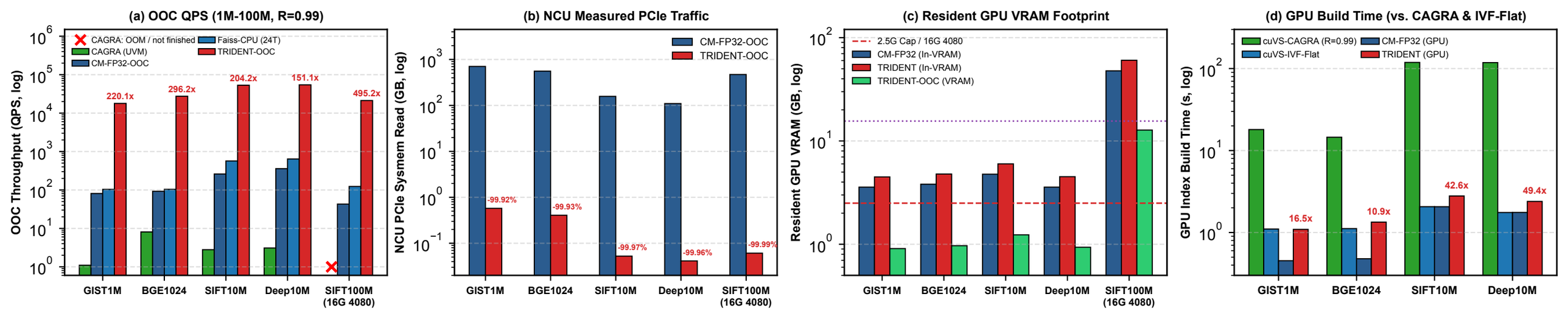
(a) OOC QPS (1M-100M, R=0.99)
OOC Throughput (QPS, log)
CAGRA: OOM / not finished
CAGRA (UVM)
CM-FP32-OOC
Faiss-CPU (24T)
TRIDENT-OOC
220.1x
296.2x
204.2x
151.1x
495.2x
GIST1M
BGE1024
SIFT10M
Deep10M
SIFT100M (16G 4080)
(b) NCU Measured PCIe Traffic
NCU PCIe Sysmem Read (GB, log)
CM-FP32-OOC
TRIDENT-OOC
-99.92%
-99.93%
-99.97%
-99.96%
-99.99%
GIST1M
BGE1024
SIFT10M
Deep10M
SIFT100M (16G 4080)
(c) Resident GPU VRAM Footprint
Resident GPU VRAM (GB, log)
2.5G Cap / 16G 4080
CM-FP32 (In-VRAM)
TRIDENT (In-VRAM)
TRIDENT-OOC (VRAM)
GIST1M
BGE1024
SIFT10M
Deep10M
SIFT100M (16G 4080)
(d) GPU Build Time (vs. CAGRA & IVF-Flat)
GPU Index Build Time (s, log)
cuVS-CAGRA (R=0.99)
cuVS-IVF-Flat
CM-FP32 (GPU)
TRIDENT (GPU)
16.5x
10.9x
42.6x
49.4x
GIST1M
BGE1024
SIFT10M
Deep10M